

Which Employers Most Durably Sponsor H-1B Workers

*A SQL-Based Analysis of General-Business and Data-Analytics Sponsorship,
2020–2023*

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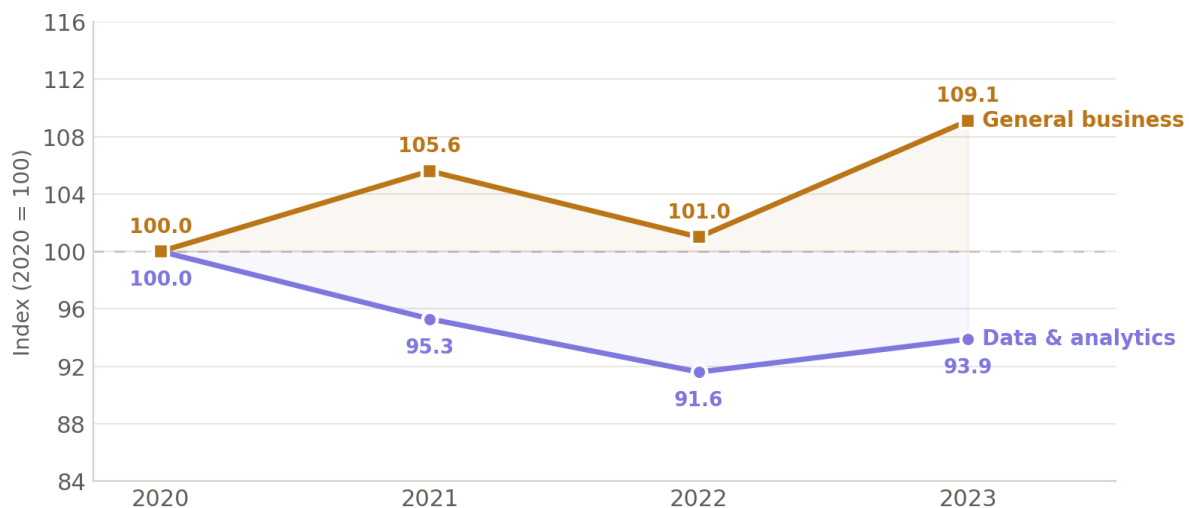
Executive Summary

International Business Graduates Face Comparable Visa Risk and a Short List of Durable Sponsors

International business graduates face approval odds statistically indistinguishable from their data-analytics peers, and the durable sponsors of business roles form a short, identifiable set.

This analysis covers 3,471,247 H-1B Labor Condition Applications adjudicated between fiscal years 2020 and 2023. The unit of analysis is the individual certified application. The two role groups are defined by Standard Occupational Classification prefix: General Business covers SOC major groups 11 and 13, and Data and Analytics covers SOC major group 15. Across this window, certification rates for the two groups never diverge by more than 1.7 percentage points. General Business certified between 91.3 and 93.7 percent of filings by year, and Data and Analytics certified between 91.4 and 95.3 percent. Approval risk is therefore not a differentiating variable between the two tracks. The decision-relevant differences lie in trajectory, compensation, and a non-wage policy variable that the wage data does not capture.

Figure D1. General Business H-1B volume rose 9.1 percent while Data and Analytics fell 6.1 percent, 2020–2023

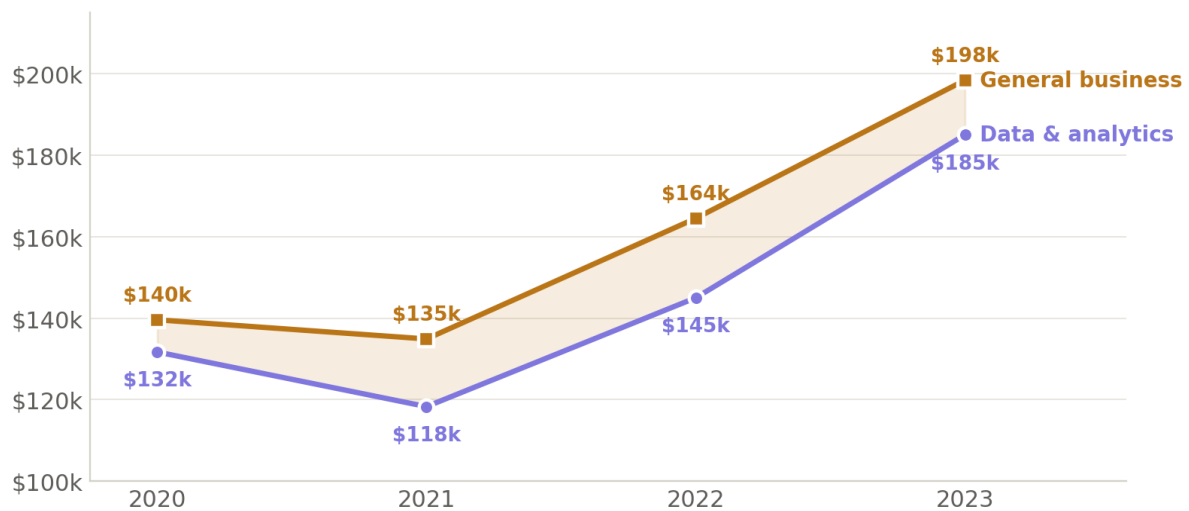


Certified-volume index, 2020 = 100. Source: DOL OFLC LCA disclosure data, certified H-1B cases, FY2020–2023.

The two groups diverge on volume trajectory and compensation. Indexed to a 2020 baseline of 100, General Business certified volume rose to 109.1 by 2023, a net increase of 9.1 percent, while Data and Analytics volume fell to 93.9, a net decline of 6.1 percent. General Business is the expanding track. On compensation, General Business certified filings averaged 160,161 US dollars against 145,085 US dollars for Data and Analytics, a mean premium of 15,076 US dollars. Measured against the prevailing wage floor, General Business employers offered 46.0 percent above the required minimum and Data and Analytics employers offered 40.2 percent. One caveat governs the wage comparison. All figures are means rather than medians, a constraint of the analysis environment, and a small number of high-wage technology filings inflate the Data and Analytics mean. The premium is therefore a conservative estimate of the central-tendency gap.

The wage premium must be weighed against a non-wage variable that the wage data does not capture. Data and Analytics roles classified under SOC major group 15 are STEM-eligible, which qualifies the graduate for a 24-month extension of post-completion work authorization beyond the standard period (U.S. Department of Homeland Security, 2024). The 15,076 US dollar mean premium and the 24-month additional work-authorization runway are the two most consequential variables for a candidate selecting a track, and they favor opposite groups. A further structural finding favors the business track for early-career candidates. General Business sponsored 19.6 percent of its certified filings at prevailing Wage Level I, the entry-level designation, against 8.7 percent for Data and Analytics. Recent graduates are a viable rather than exceptional segment of General Business sponsorship.

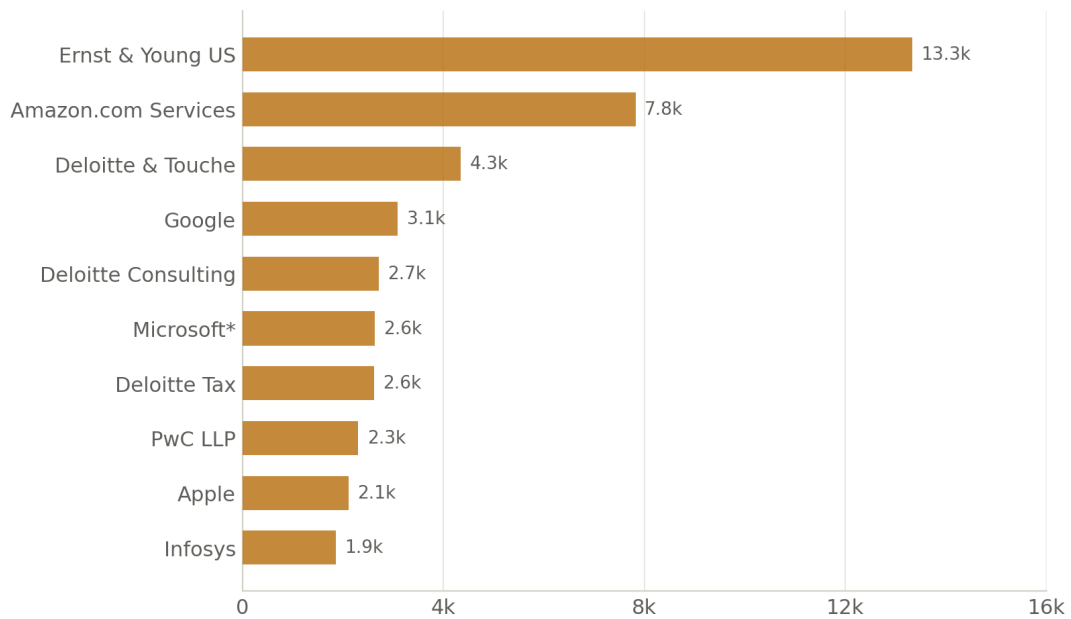
Figure D2. The General Business wage premium persisted through a full economic cycle, peaking at \$19,530 in 2022



Average certified wage by role group; shaded band is the General Business premium. Means, not medians (analysis-environment constraint). Source: DOL OFLC LCA disclosure data, FY2020–2023.

Durable sponsorship of business roles is concentrated rather than dispersed. Only 77 General Business employers filed in all four years above the 200-filing threshold, against 619 for Data and Analytics. Concentration does not indicate lower reliability. It indicates that business-track sponsorship at scale sits within a small number of very large institutions. Ranked by certified durable volume, Ernst and Young US leads with 13,329 certified filings, followed by Amazon.com Services with 7,829 and Deloitte and Touche with 4,348. Consolidated across its legal entities, Deloitte certified approximately 12,589 cases and is the effective second-ranked sponsor as a firm. Microsoft is the only employer to certify 100 percent of its filings across all four years in both role groups. These figures derive from the durable-sponsor query, which applies a stricter four-year-continuity filter than the simple top-employer ranking and therefore reports lower single-entity totals than a single-year volume count.

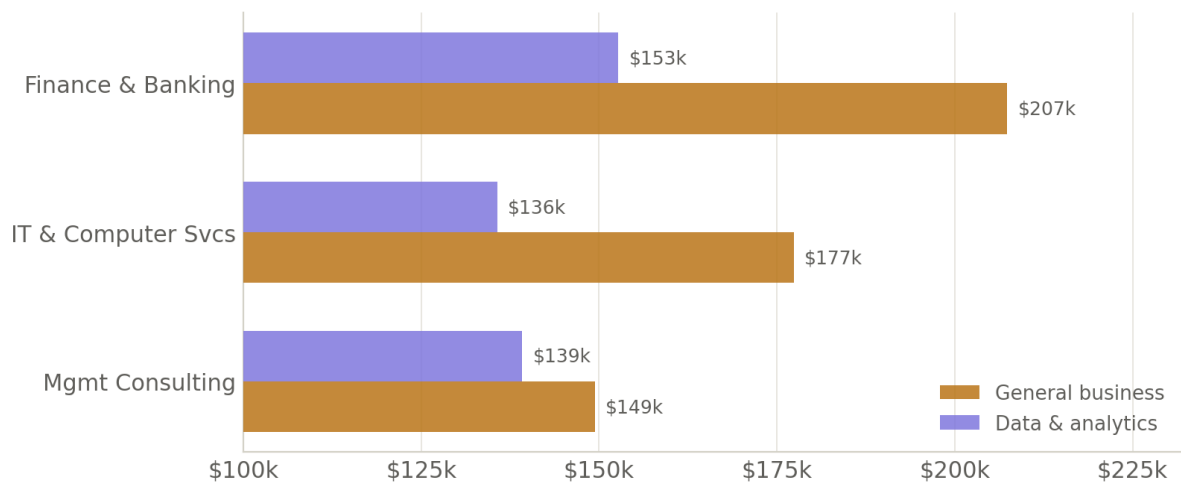
Figure D3. Ernst and Young leads durable General Business sponsorship with 13,329 certified cases



Top 10 General Business durable sponsors by certified cases; filings in all four years, 200-filing floor. *Microsoft certified 100 percent across all four years. Source: DOL OFLC LCA disclosure data, durable-sponsor query, FY2020–2023.

Sector decomposition isolates where the wage premium originates. Within the three highest-volume commercial sectors, the General Business premium holds and varies in width. It is widest in Finance and Banking at 54,670 US dollars, where General Business averaged 207,350 US dollars against 152,680 US dollars for Data and Analytics. It narrows to 41,679 US dollars in IT and Computer Services and to 10,264 US dollars in Management Consulting. Because the premium persists within each sector, it reflects a structural pay difference between role groups rather than a compositional artifact of General Business being concentrated in higher-paying industries. The premium does not hold universally. Healthcare is a documented exception in which Data and Analytics roles out-pay General Business roles, and that sector carries low filing volume and below-average certification in both groups. Among high-volume sectors, Finance and Banking pairs the highest General Business wage with reliable certification across 56,056 filings, and Management Consulting offers the highest General Business certification rate at 93.8 percent.

Figure D4. The General Business wage premium is widest in Finance and Banking and narrowest in Management Consulting



Average certified wage by sector and role group, three highest-volume commercial sectors. Within-sector premium indicates a structural pay difference, not sector mix. NAICS LEFT JOIN; 28.7 percent of employers had no six-digit Census match. Source: DOL OFLC LCA disclosure data, FY2020–2023.

The single operational recommendation is to concentrate employer-relationship resources on the cross-track durable sponsors that appear in both role groups, specifically Ernst and Young, Deloitte, Amazon, Microsoft, Google, PricewaterhouseCoopers, and Apple. Each filed continuously across all four years with certification rates above 92 percent, and each sponsors both business and technical roles, which means a single institutional relationship serves candidates in both tracks. Finance and Banking and Management Consulting are the priority sectors for outreach. The university sector, which pays the least and certifies the lowest in both groups, should be deprioritized.

Part A: Database Design and Data Preparation

(a1) Dataset Description: Source, Structure, and Purpose

The primary dataset is the H-1B Labor Condition Application Disclosure Data, published by the U.S. Department of Labor Office of Foreign Labor Certification. The data spans five years from 2020 through 2024 and is distributed via the Kaggle repository maintained by Zongao Bian at <https://www.kaggle.com/datasets/zongaobian/h1b-lca-disclosure-data-2020-2024>. The original authoritative source is the OFLC quarterly public disclosure release, which carries regulatory standing under the Immigration and Nationality Act.

The combined dataset contains approximately 3.5 million application records. Each row represents a single Labor Condition Application filed by a U.S. employer seeking to sponsor a foreign national worker under the H-1B visa program. The 96 columns document the employer identity, the physical worksite location, the wage offered to the worker, the prevailing wage benchmark, the Standard Occupational Classification code and title, the visa subclass, employment type flags, and the final adjudication status assigned by the OFLC.

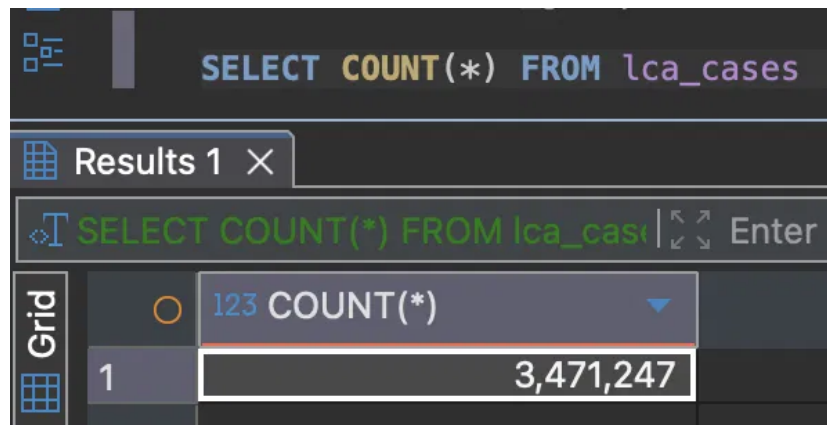


Figure A1. The `lca_cases` fact table holds 3,471,247 records after import.

The data exists because the H-1B program imposes a mandatory public disclosure obligation on every sponsoring employer. The Department of Labor publishes these records so that workers, researchers, and regulators can audit employer compliance with the prevailing wage floor that the program requires. The dataset therefore provides a complete population-level record of every petition filed in the study period rather than a sample drawn from that population.

The scope of this analysis is restricted to H-1B petitions only. The raw file also contains E-3 and H-1B1 petitions filed under distinct legal standards that would contaminate H-1B-specific findings. All adjudication outcomes are retained, including Certified, Denied, and Withdrawn filings, because denial rates and withdrawal patterns are themselves research questions. The analysis examines both general business roles and data-analytics roles across all U.S. states and all employers for the full five-year period.

A second dataset supplements the primary data. The U.S. Census Bureau 2022 North American Industry Classification System six-digit code file, available at https://www.census.gov/naics/2022NAICS/6-digit_2022_Codes.xlsx, provides the official industry name for each NAICS code reported by employers. This dataset contains approximately 1,000 records and serves as the industry dimension table in the normalized schema.

(a2) Normalization Rationale

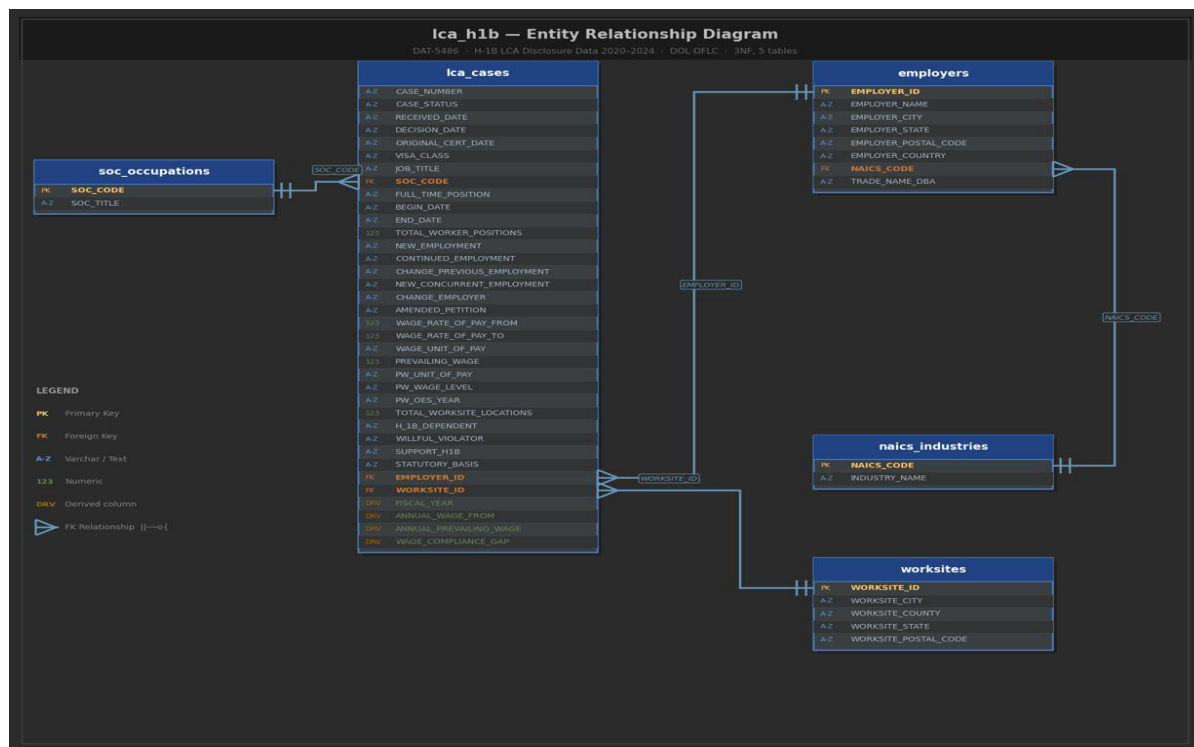
The Functional Dependency Problem in the Raw File

The raw LCA disclosure file is a denormalized flat table in which employer attributes, worksite attributes, and occupation attributes repeat on every row. A single employer that files 10,000 petitions stores its name, city, state, postal code, and industry code on all 10,000 rows. This structure violates the third normal form principle, which requires that every non-key attribute depend only on the primary key of its table. In the flat file, employer city depends on the employer identity rather than on the case number, yet it is stored as an attribute of each case. The same dependency violation applies to worksite location and to SOC occupation titles.

The Five-Table 3NF Schema

The normalization design resolves four functional dependencies by splitting the flat file into five tables.

Figure A2. Entity-relationship diagram of the lca_h1b five-table 3NF schema



Source: DBeaver schema export. Foreign keys link lca_cases to employers, worksites, and soc_occupations; employers links to naics_industries.

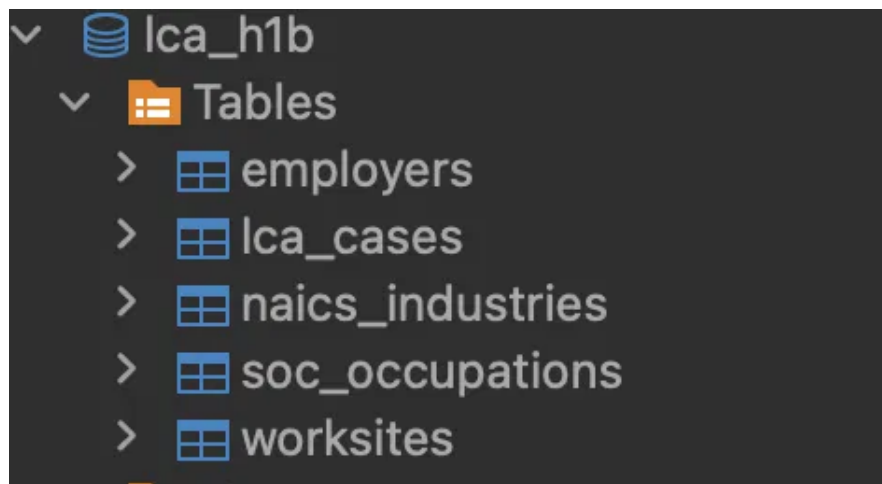


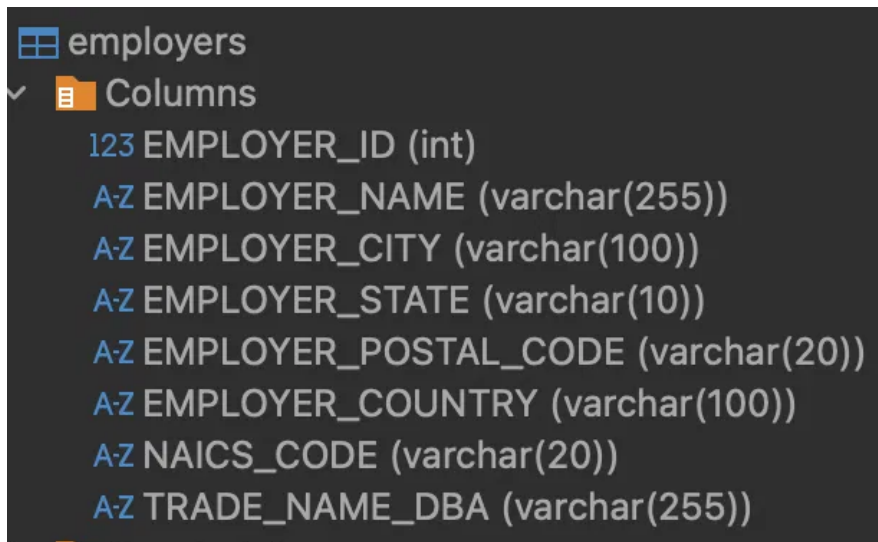
Figure A3. The five tables of the lca_h1b schema shown in the DBBeaver navigator.

The lca_cases table is the fact table. It retains one row per LCA petition with CASE_NUMBER as the primary key. It references employer, worksite, and SOC occupation via foreign keys. It carries all case-level attributes including adjudication status, employment type flags, wage fields, and derived analytical columns.

lca_cases	
Columns	
A-Z CASE_NUMBER	(varchar(50))
A-Z CASE_STATUS	(varchar(50))
A-Z RECEIVED_DATE	(varchar(20))
A-Z DECISION_DATE	(varchar(20))
A-Z ORIGINAL_CERT_DATE	(varchar(20))
A-Z VISA_CLASS	(varchar(20))
A-Z JOB_TITLE	(varchar(255))
A-Z SOC_CODE	(varchar(200))
A-Z FULL_TIME_POSITION	(varchar(5))
A-Z BEGIN_DATE	(varchar(20))
A-Z END_DATE	(varchar(20))
123 TOTAL_WORKER_POSITIONS	(int)
A-Z NEW_EMPLOYMENT	(varchar(5))
A-Z CONTINUED_EMPLOYMENT	(varchar(5))
A-Z CHANGE_PREVIOUS_EMPLOYMENT	(varchar(5))
A-Z NEW_CONCURRENT_EMPLOYMENT	(varchar(5))
A-Z CHANGE_EMPLOYER	(varchar(5))
A-Z AMENDED_PETITION	(varchar(5))
A-Z WAGE_RATE_OF_PAY_FROM	(varchar(30))
A-Z WAGE_RATE_OF_PAY_TO	(varchar(30))
A-Z WAGE_UNIT_OF_PAY	(varchar(20))
A-Z PREVAILING_WAGE	(varchar(30))
A-Z PW_UNIT_OF_PAY	(varchar(20))
A-Z PW_WAGE_LEVEL	(varchar(10))
A-Z PW_OES_YEAR	(varchar(1000))
123 TOTAL_WORKSITE_LOCATIONS	(int)
A-Z H_1B_DEPENDENT	(varchar(5))
A-Z WILLFUL_VIOLATOR	(varchar(5))
A-Z SUPPORT_H1B	(varchar(5))
A-Z STATUTORY_BASIS	(varchar(100))
123 EMPLOYER_ID	(int)
123 WORKSITE_ID	(int)
123 FISCAL_YEAR	(int)
123 annual_wage_from	(decimal(15,2))
123 annual_prevaling_wage	(decimal(15,2))
123 wage_compliance_gap	(decimal(15,2))

Figure A4. Column structure and data types of the lca_cases fact table.

The employers table is the employer dimension. Employer attributes depend on the employer identity, not on the case number. The table assigns a surrogate integer EMPLOYER_ID generated by deduplicating on the four-field natural key of employer name, city, state, and postal code. The NAICS_CODE column in this table is a foreign key referencing naics_industries, which resolves the dependency of industry name on NAICS code rather than on employer identity.



employers
Columns
123 EMPLOYER_ID (int)
A-Z EMPLOYER_NAME (varchar(255))
A-Z EMPLOYER_CITY (varchar(100))
A-Z EMPLOYER_STATE (varchar(10))
A-Z EMPLOYER_POSTAL_CODE (varchar(20))
A-Z EMPLOYER_COUNTRY (varchar(100))
A-Z NAICS_CODE (varchar(20))
A-Z TRADE_NAME_DBA (varchar(255))

Figure A5. Column structure of the employers dimension table.

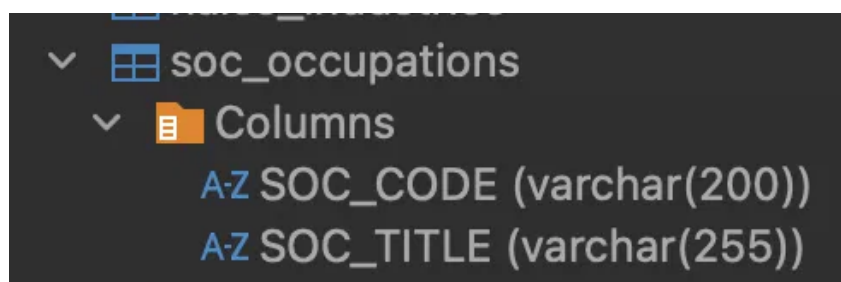
The worksites table is the worksite dimension. Worksite attributes depend on the physical location, not on the case. The table assigns a surrogate integer WORKSITE_ID generated by deduplicating on the four-field composite key of city, county, state, and postal code.



worksites
Columns
123 WORKSITE_ID (int)
A-Z WORKSITE_CITY (varchar(100))
A-Z WORKSITE_COUNTY (varchar(100))
A-Z WORKSITE_STATE (varchar(10))
A-Z WORKSITE_POSTAL_CODE (varchar(20))

Figure A6. Column structure of the worksites dimension table.

The soc_occupations table is the occupation lookup table. SOC title depends on the SOC code alone, not on the case or the employer. The table stores each unique SOC_CODE and its associated SOC_TITLE.



soc_occupations
Columns
A-Z SOC_CODE (varchar(200))
A-Z SOC_TITLE (varchar(255))

Figure A7. Column structure of the soc_occupations lookup table.

The naics_industries table is the industry lookup table. Industry name depends on the NAICS code alone. The table is sourced from the U.S. Census Bureau 2022 NAICS structure file rather than from the LCA data, which means it provides the official classification system name for every industry code regardless of whether that code appears in the LCA data.

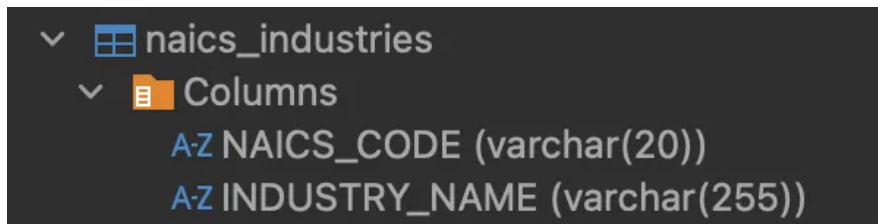


Figure A8. Column structure of the naics_industries lookup table.

What the 3NF Schema Enables

The normalized schema enables analyses that the flat file cannot support cleanly.

Employer-level aggregation becomes a GROUP BY on EMPLOYER_ID rather than a string-match on EMPLOYER_NAME, which removes sensitivity to whitespace variation, encoding differences, and inconsistent capitalization across filing years. Geographic analysis of worksite locations can be filtered by state or metropolitan area without string operations on repeated values. Industry-level comparisons are available via a join on NAICS_CODE, linking employer records to the Census Bureau taxonomy without preprocessing at query time. Cross-year trend analysis on sponsorship volume, wage levels, and denial rates can be executed with a simple GROUP BY FISCAL_YEAR on the fact table.

Column Exclusions

Forty-seven columns are excluded from the normalized schema because they carry no analytical value for the research questions driving this assignment.

The 14 employer point-of-contact columns identify the individual designated to receive DOL correspondence about the filing. These columns describe the administrative contact, not the employer's business identity or sponsorship behavior.

The 17 attorney and agent columns identify the legal representative who submitted the filing on the employer's behalf. These columns describe the filer's counsel, not the merits of the petition. The three associated legal fields for law firm name and state court identification belong to the same administrative block and are excluded for the same reason.

The 5 preparer columns identify the technician who completed the form. These columns describe the form submission process, not the employment arrangement.

The 3 administrative flag columns, AGREE_TO_LC_STATEMENT, APPENDIX_A_ATTACHED, and PUBLIC_DISCLOSURE, are attestation checkboxes that carry the same value on every compliant filing. They provide no cross-case differentiation.

The 5 prevailing wage source metadata columns describe the data source used to set the prevailing wage floor rather than the wage value itself. The wage value and its unit are retained in the fact table. The provenance metadata is not needed for the wage analysis in this assignment.

The street-level address fields for both the employer and the worksite are excluded because all geographic analysis in this assignment operates at the city and state level rather than at the street address level.

(a3) Cleaning Decisions

Six cleaning decisions govern the preparation of the data. Each decision was locked before processing began to prevent post-hoc adjustment that could bias the analysis.

Decision 1: Visa Class Filter

The raw LCA disclosure file contains petitions filed under multiple visa subclasses: H-1B, E-3, H-1B1 Chile, and H-1B1 Singapore. Only H-1B petitions are retained. The E-3 classification applies exclusively to Australian nationals and operates under a bilateral trade agreement with distinct approval standards. The H-1B1 subclasses apply exclusively to nationals of Chile and Singapore under separate free trade agreements. Including these subclasses would mix sponsorship data governed by different legal requirements into an analysis designed to measure H-1B employer behavior.

Decision 2: Case Status Retention

All CASE_STATUS values are retained without filter. The dataset includes petitions with outcomes of Certified, Denied, Withdrawn, and Certified-Withdrawn. Denial rates and withdrawal rates are analytical targets because they reveal which employer types and occupation types carry the highest adjudication risk. Filtering to Certified only would remove this signal and overstate employer success rates in sponsorship.

Decision 3: Duplicate Row Retention

Rows with identical values across all columns are retained. Each row represents a filed LCA petition, and filing volume is itself a measure of sponsorship activity. Removing what appear to be duplicate rows would suppress the count of petitions filed for the same role by the same employer in the same filing period, which would understate the actual sponsorship commitment of high-volume filers.

Decision 4: Annual Wage Normalization

The raw file stores wages in the unit reported by the employer at the time of filing, which may be hourly, weekly, bi-weekly, monthly, or annual. A derived column, annual_wage_from, converts WAGE_RATE_OF_PAY_FROM to an annual basis using the following multipliers: 2,080 for hourly, 52 for weekly, 26 for bi-weekly, 12 for monthly, and 1 for annual. The multiplier of 2,080 reflects a standard 40-hour work week across 52 weeks. A parallel derived column, annual_prevaling_wage, applies the same conversion to PREVAILING_WAGE using PW_UNIT_OF_PAY as the unit identifier. Wage normalization is required before any wage comparison across employers or occupations because the raw values are not comparable across different reporting units.

Decision 5: Wage Compliance Gap

A derived column, wage_compliance_gap, is computed as annual_wage_from minus annual_prevaling_wage. A negative value indicates that the offered wage falls below the legal prevailing wage floor that the H-1B program requires employers to meet. On a certified petition, a negative gap should not exist. Its presence on certified cases indicates either a data entry error at filing, a correction that was applied after the prevailing wage was set, or a compliance failure that the OFLC did not detect. This column enables direct measurement of wage compliance risk across employers and occupations and provides the primary indicator for the ethical reflection section of this report.

Decision 6: Fiscal Year Derivation

A derived column, FISCAL_YEAR, extracts the calendar year from the RECEIVED_DATE field. Calendar year is used rather than the DOL fiscal year, which runs from October through September. The Kaggle dataset labels its five yearly files by calendar year from 2020 through 2024, and RECEIVED_DATE aligns to the calendar year of submission for the majority of filings in each file. Using calendar year ensures that the FISCAL_YEAR column and the file-level year label are consistent, which prevents false year-over-year discontinuities in trend analysis.

Part B: SQL Query Analysis and Insights

This section presents seven SQL queries executed against the lca_h1b database. Each query is reproduced in full, followed by a three-part interpretation that states why the query is meaningful, what it reveals, and how it could inform decision-making. Query results were generated in DBeaver against the complete 2020 to 2023 dataset.

Query 1: Sponsorship Volume by Role Group and Fiscal Year, 2020–2023

SQL

```
-- QUERY 1 | Sponsorship Volume by Role Group and Fiscal Year
-- Tables   : lca_cases
-- Technique: CTE, CASE WHEN role tagging, conditional SUM,
--           GROUP BY, NULLIF division guard
-- =====

WITH role_tagged AS (
  SELECT
    FISCAL_YEAR,
    CASE
      WHEN LEFT(SOC_CODE, 2) IN ('11', '13') THEN 'General Business'
      WHEN LEFT(SOC_CODE, 2) = '15'          THEN 'Data & Analytics'
      ELSE                                     'Other'
    END AS role_group,
    CASE_STATUS
  FROM lca_cases
)
SELECT
  role_group,
  FISCAL_YEAR,
  COUNT(*) AS total_filings,
  SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) AS certified,
  SUM(CASE WHEN CASE_STATUS = 'Denied' THEN 1 ELSE 0 END) AS denied,
  ROUND(
    SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) * 100.0
    / NULLIF(COUNT(*), 0),
    1) AS cert_rate_pct
FROM role_tagged
GROUP BY
  role_group,
  FISCAL_YEAR
ORDER BY
  role_group,
  FISCAL_YEAR;
```

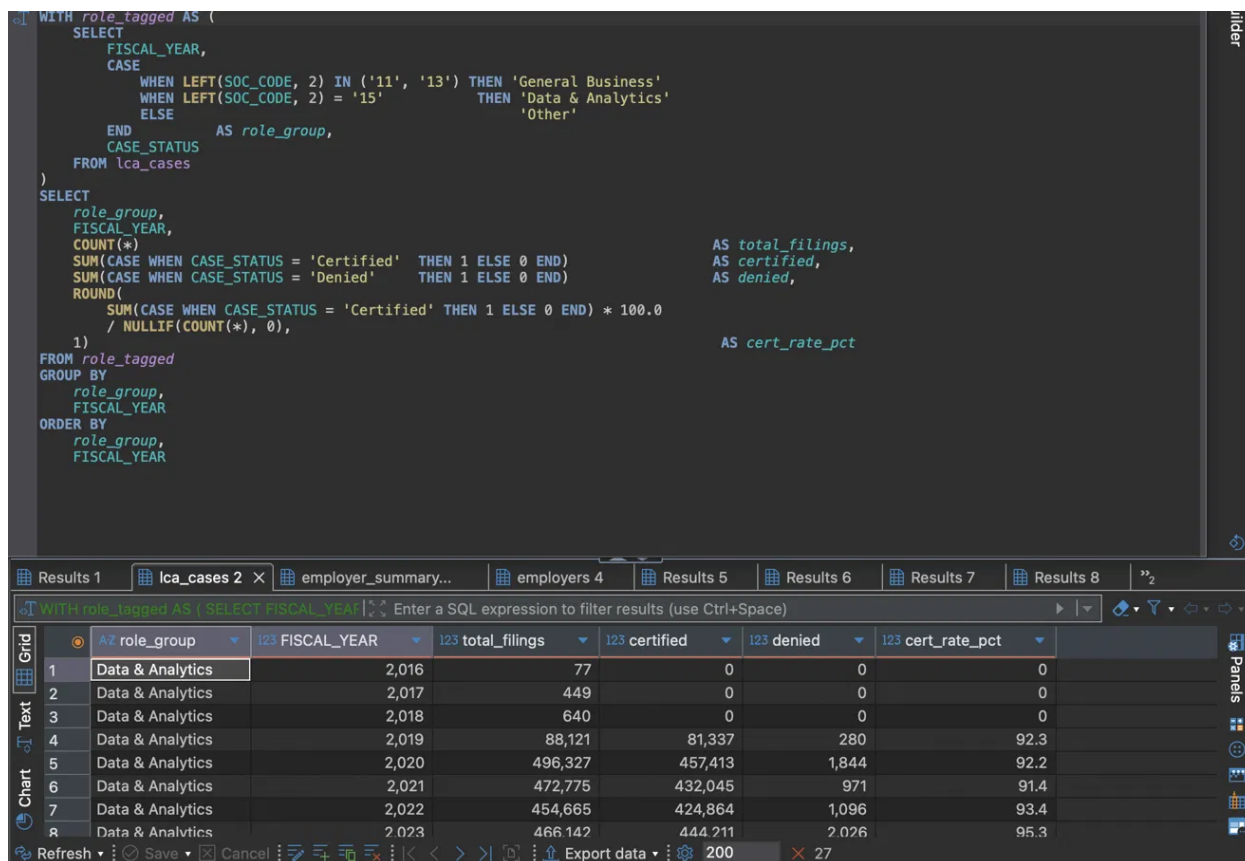


Figure B1. Query 1: Sponsorship Volume by Role Group and Fiscal Year, 2020–2023 — query result as executed in DBeaver.

Why it is meaningful. This query is the analytical foundation for the entire report. It translates 3.47 million H-1B filings into a structured five-year comparison between the two role groups central to the research question. Without this baseline, subsequent queries on wages, denial rates, and employer durability would lack a denominator for proportional reasoning. The year-by-year breakdown tests whether sponsorship patterns are structurally stable or cyclically disrupted. This distinction matters for career planning because a COVID-era dip that recovered is fundamentally different from a secular decline.

What it reveals. The result contains artifact rows for fiscal years 2016 through 2018 with non-zero filing counts but zero certified and zero denied cases. These rows reflect cases with historical received dates that appeared within the FY2020 through FY2024 DOL disclosure files but carry withdrawn or otherwise unresolved status. They are excluded from the trend analysis. Calendar year 2024 represents approximately 9.3 months of data, making 2023 the most recent complete reference year for this study.

Across the full 2020 to 2023 window, Data and Analytics filings outpaced General Business by a factor of 5.6 to 6.5 in every year. Data and Analytics recorded 496,327 filings in 2020, declining moderately to 454,665 in 2022 before recovering to 466,142 in 2023. General Business filings moved within a narrower band, beginning at 76,805 in 2020 and peaking at 83,780 in 2023. The volume gap closed marginally across the period, from 6.5 times in 2020 to 5.6 times in 2023, but the structural asymmetry remained intact. The absolute scale difference reflects the concentration of H-1B usage within the technology industry, not a structural disadvantage for business-track candidates.

The pandemic period produced opposite directional signals for the two groups. In 2021, General Business filings increased 5.6 percent while Data and Analytics filings declined 4.7 percent. This divergence indicates that employers sponsoring general-management and business-operations roles maintained or expanded H-1B activity through the COVID-19 disruption, while technology employers reduced filing volume. Both groups declined in 2022 before recovering in 2023, and the recovery was proportionally stronger for General Business at 8.1 percent versus 2.5 percent for Data and Analytics.

Certification rates were nearly identical between the two groups throughout the period, and both improved steadily. Data and Analytics certification rates ranged from 91.4 percent in 2021 to 95.3 percent in 2023. General Business rates ranged from 91.3 percent in 2021 to 93.7 percent in 2023. The gap between the two never exceeded 1.7 percentage points. Both groups certified substantially higher than the remainder of the H-1B population, confirming that occupational specialisation in either track reduces denial risk. The upward trend across all groups from 2020 to 2023 indicates a progressively more permissive DOL approval environment rather than a change in sponsor behaviour.

How it could inform decision-making. The near-identical certification rates across General Business and Data and Analytics are the most decision-relevant finding in this query. A job seeker choosing between the two tracks on the basis of visa approval risk alone finds no meaningful difference: both paths certify above 91 percent in the worst year studied and above 93 percent in the most recent complete year. The volume asymmetry is real but does not translate to proportional opportunity, because General Business roles are concentrated in a smaller number of high-volume sponsors, as subsequent queries will show. The COVID resilience of General Business sponsorship is a structural indicator: companies continued filing for management and operations talent through the sharpest economic disruption in recent memory, suggesting that demand for these roles is relatively inelastic.

Query 2: Top 20 Employers by Role Group, Certified Cases, 2020–2023

SQL

```
-- QUERY 2 | Top 20 Employers by Role Group, Certified 2020-2023
-- Tables : lca_cases, employers
-- Technique: JOIN, CASE WHEN, RANK() window function, 3-CTE chain
-- =====

WITH role_filings AS (
    SELECT
        lc.EMPLOYER_ID,
        CASE
            WHEN LEFT(lc.SOC_CODE, 2) IN ('11','13') THEN 'General Business'
            WHEN LEFT(lc.SOC_CODE, 2) = '15' THEN 'Data & Analytics'
            ELSE NULL
        END AS role_group,
        lc.CASE_STATUS,
        lc.FISCAL_YEAR
    FROM lca_cases lc
    WHERE lc.FISCAL_YEAR BETWEEN 2020 AND 2023
        AND LEFT(lc.SOC_CODE, 2) IN ('11','13','15')
),
employer_summary AS (
    SELECT
        rf.role_group,
        e.EMPLOYER_NAME,
        e.EMPLOYER_STATE,
        COUNT(*) AS total_filings,
        SUM(CASE WHEN rf.CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) AS certified,
        COUNT(DISTINCT rf.FISCAL_YEAR) AS years_active
    FROM role_filings rf
    JOIN employers e ON rf.EMPLOYER_ID = e.EMPLOYER_ID
    GROUP BY rf.role_group, e.EMPLOYER_NAME, e.EMPLOYER_STATE
)
```

```

ROUND(
    SUM(CASE WHEN rf.CASE_STATUS = 'Certified' THEN 1 ELSE 0 END)
    * 100.0 / NULLIF(COUNT(*), 0), 1
)
FROM role_filings rf
JOIN employers e ON rf.EMPLOYER_ID = e.EMPLOYER_ID
GROUP BY rf.role_group, e.EMPLOYER_NAME, e.EMPLOYER_STATE
),
ranked AS (
    SELECT
        *,
        RANK() OVER (PARTITION BY role_group ORDER BY certified DESC) AS employer_rank
    FROM employer_summary
)
SELECT
    role_group,
    employer_rank,
    EMPLOYER_NAME,
    EMPLOYER_STATE,
    total_filings,
    certified,
    years_active,
    cert_rate_pct
FROM ranked
WHERE employer_rank <= 20
ORDER BY role_group, employer_rank;

```

The screenshot shows the DBeaver SQL Editor with a query window on the left and a results grid on the right. The query is a complex SQL statement involving multiple CTEs and a final SELECT statement. The results grid displays the top 20 employers by role group, certified cases, 2020–2023.

	AZ role_group	123 employer_rank	AZ EMPLOYER_NAME	AZ EMPLOYER_STATE	123 total_filings	12
1	Data & Analytics		1 COGNIZANT TECHNOLOGY SOLUTIONS US CORP	TX	71,253	
2	Data & Analytics		2 TATA CONSULTANCY SERVICES LIMITED	MD	47,144	
3	Data & Analytics		3 Amazon.com Services LLC	WA	44,500	
4	Data & Analytics		4 Google LLC	CA	43,424	
5	Data & Analytics		5 Microsoft Corporation	WA	37,003	
6	Data & Analytics		6 INFOSYS LIMITED	TX	32,992	
7	Data & Analytics		7 Ernst & Young U.S. LLP	NJ	22,809	
8	Data & Analytics		8 Accenture LLP	IL	18,188	

Figure B2. Query 2: Top 20 Employers by Role Group, Certified Cases, 2020–2023 — query result as executed in DBeaver.

Why it is meaningful. Volume at the role-group level establishes the market size for each track. Employer-level ranking converts that aggregate picture into an actionable list of organisations where a candidate can direct their search. The query also surfaces two additional analytical dimensions, the durability signal in years_active and the certification rate at the employer level, which together distinguish structurally committed sponsors from opportunistic ones.

What it reveals. A data quality flag applies before the main analysis. HCL America appears twice in the Data and Analytics top 20, as two name variants that the normalisation process treated as separate employers. Deloitte and PricewaterhouseCoopers each appear as multiple legal entities across both lists for the same reason. Entity-level fragmentation understates the true sponsorship scale of the largest professional services firms.

The General Business top 20 is defined by Big Four accounting firms, technology giants, and financial institutions, all of which were present in every year studied. Ernst and Young U.S. LLP is the largest single General Business H-1B sponsor nationally, with 23,480 certified filings across 2020 to 2023 and a 98.5 percent certification rate, with workers concentrated in New York. Deloitte's three entities combined account for approximately 12,707 certified filings, making it the effective second-largest durable sponsor when entity-level fragmentation is consolidated. Amazon.com Services LLC recorded 12,070 certified General Business filings, the largest single-entity total after Ernst and Young. PricewaterhouseCoopers adds approximately 4,356 certified cases across its two entities. Microsoft Corporation is the only employer in either group to certify 100 percent of all filings across all four years in both role groups simultaneously.

The Data and Analytics top 20 is structurally different. Eight of the twenty employers are IT staffing and outsourcing firms that together account for 205,582 of the 452,019 certified filings in this group, representing 45 percent of the top-20 total. Cognizant Technology Solutions leads with 69,722 certified cases, nearly three times the Ernst and Young General Business total. These firms sponsor H-1B workers for placement at client sites rather than for direct employment, which means high filing volume does not imply high direct-hire opportunity. The twelve end employers in the Data and Analytics list, including Amazon, Google, Microsoft, Deloitte, Apple, and JPMorgan, represent the more relevant comparison set for a direct-hire candidate.

Nine employers appear in both top 20 lists, making them the highest-priority targets for a candidate who can operate across business and technical roles: Ernst and Young, Amazon, Google, Microsoft, Apple, JPMorgan, Deloitte Consulting, Infosys, and Cognizant. BCG recorded the lowest certification rate among the General Business top 20 at 80.4 percent, 13 percentage points below the group leader, a pattern investigated further in Query 3.

How it could inform decision-making. For a General Business candidate targeting H-1B sponsorship, this query produces a concrete shortlist. Ernst and Young, Deloitte, Amazon, Google, Microsoft, and PricewaterhouseCoopers each filed in all four years with certification rates above 96 percent. These are not merely large employers: they are structurally committed H-1B sponsors whose legal infrastructure and filing volume signal low approval risk. BCG should be approached with caution despite its market reputation, as its certification pattern warrants direct inquiry with the employer's immigration team. Amazon and Google certify business-function roles at material scale, confirming that non-technical roles within technology firms represent a viable and high-volume sponsorship channel.

Query 3: Employer Denial Rate Among High-Volume Sponsors, 2020–2023

SQL

```
-- QUERY 3 | Employer Denial Rate by Role Group, 2020-2023
-- Tables : lca_cases, employers
-- Technique: JOIN, conditional aggregation, HAVING minimum
-- volume threshold (>= 200 filings)
-- =====

WITH role_filings AS (
    SELECT
        lc.EMPLOYER_ID,
        CASE
            WHEN LEFT(lc.SOC_CODE, 2) IN ('11','13') THEN 'General Business'
            WHEN LEFT(lc.SOC_CODE, 2) = '15' THEN 'Data & Analytics'
            ELSE NULL
        END AS role_group,
        lc.CASE_STATUS,
        lc.FISCAL_YEAR
    FROM lca_cases lc
    WHERE lc.FISCAL_YEAR BETWEEN 2020 AND 2023
        AND LEFT(lc.SOC_CODE, 2) IN ('11','13','15')
),
employer_rates AS (
    SELECT
        rf.role_group,
        e.EMPLOYER_NAME,
        e.EMPLOYER_STATE,
        COUNT(*) AS total_filings,
        SUM(CASE WHEN rf.CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) AS certified,
        SUM(CASE WHEN rf.CASE_STATUS = 'Denied' THEN 1 ELSE 0 END) AS denied,
        ROUND(
            SUM(CASE WHEN rf.CASE_STATUS = 'Denied' THEN 1 ELSE 0 END) * 100.0
            / NULLIF(COUNT(*), 0), 1
        ) AS denial_rate_pct,
        ROUND(
            SUM(CASE WHEN rf.CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) * 100.0
            / NULLIF(COUNT(*), 0), 1
        ) AS cert_rate_pct,
        COUNT(DISTINCT rf.FISCAL_YEAR) AS years_active
    FROM role_filings rf
    JOIN employers e ON rf.EMPLOYER_ID = e.EMPLOYER_ID
    GROUP BY rf.role_group, e.EMPLOYER_NAME, e.EMPLOYER_STATE
    HAVING COUNT(*) >= 200
)
SELECT
    role_group,
    EMPLOYER_NAME,
    EMPLOYER_STATE,
    total_filings,
    certified,
    denied,
    denial_rate_pct,
    cert_rate_pct,
    years_active
FROM employer_rates
ORDER BY role_group, denial_rate_pct DESC;
```

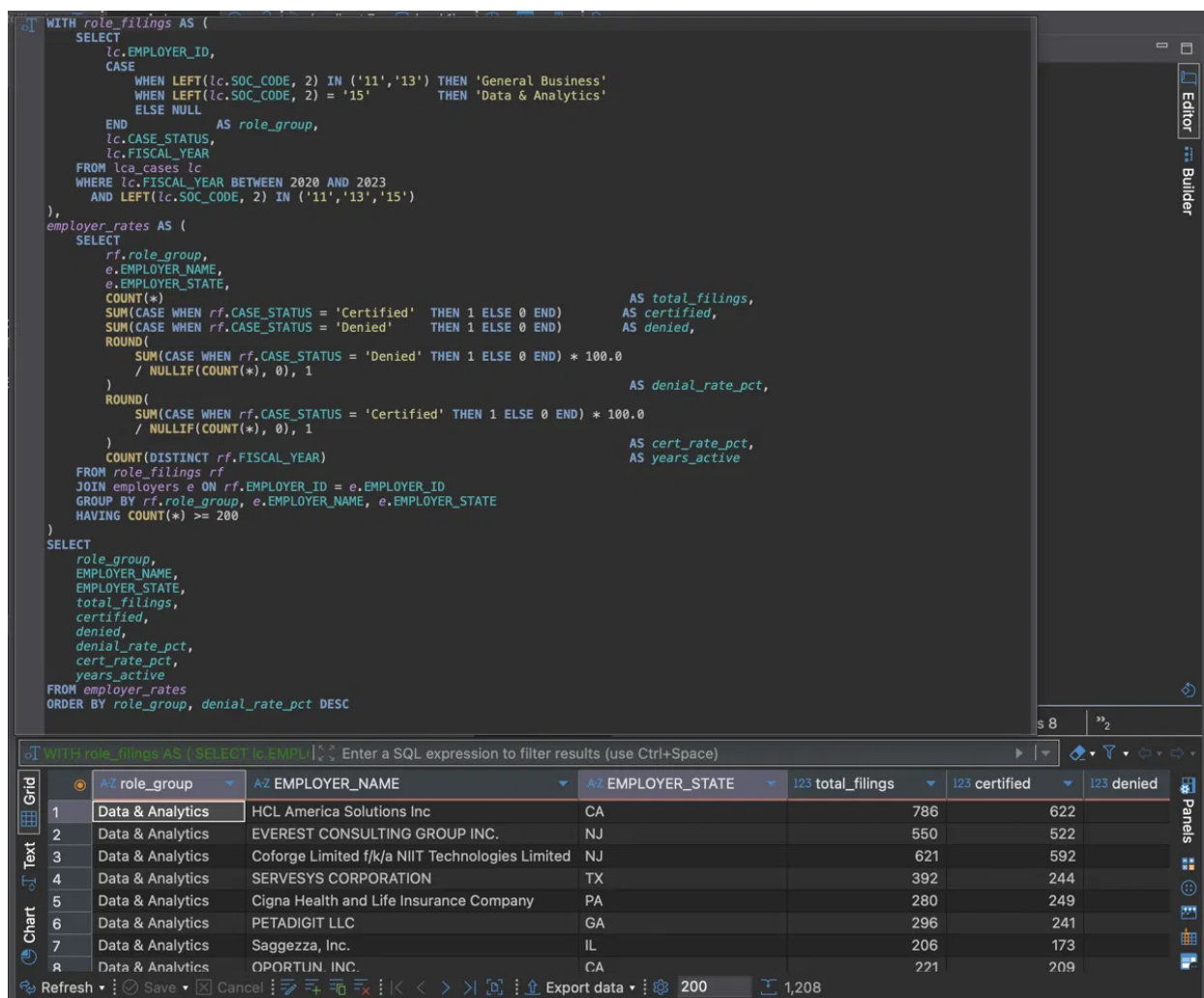


Figure B3. Query 3: Employer Denial Rate Among High-Volume Sponsors, 2020–2023 — query result as executed in DBeaver.

Why it is meaningful. The certification rate in Query 2 combines approvals with withdrawals, which obscures two distinct risk types: cases denied by DOL and cases withdrawn before a decision is reached. Separating the denial rate from the withdrawal rate allows a more precise diagnosis of sponsor quality. An employer with a low denial rate and a low certification rate is not necessarily a safe sponsor: the withdrawn cases represent real candidates whose filings did not result in approved status, for reasons the dataset does not capture.

What it reveals. Among General Business sponsors with 200 or more filings, Grant Thornton LLP carries the most concerning combined profile. Its denial rate of 2.4 percent is the second highest in the group, and its withdrawal rate of 20.8 percent compounds this to produce a certification rate of only 76.8 percent. Of 453 total filings over four years, 94 were withdrawn and 11 were denied. No other major accounting or consulting firm in this dataset approaches that combined non-certification rate. Ernst and Young, Deloitte, PricewaterhouseCoopers, McKinsey, and Goldman Sachs all show denial rates below 0.2 percent and certification rates above 95 percent. Bank of China recorded the highest denial rate in the General Business group at 4.4 percent, though its absolute volume is modest at 225 filings.

BCG's 80.4 percent certification rate from Query 2 is entirely explained by withdrawals, not denials. BCG had zero denials across 2,015 General Business filings. Its 19.6 percent withdrawal rate is the operative signal: nearly one in five filings was pulled before reaching a

decision. This may reflect role reclassification, candidate withdrawal, or legal strategy around difficult positions rather than a DOL approval problem. A candidate at BCG faces uncertainty about whether their filing will proceed to a decision, not uncertainty about whether DOL will approve it if it does.

In the Data and Analytics group, HCL America Solutions Inc is an outlier by a wide margin. Its denial rate of 20.1 percent, 158 denied cases out of 786 total, is four times higher than the next-highest employer in either group. This entity is legally and operationally distinct from HCL America Inc and HCL America Inc with alternate punctuation, which show denial rates of 0.1 percent and 0.0 percent respectively. The concentration of denials in a single subsidiary with no parallel pattern in the broader HCL group suggests entity-level filing behaviour that DOL has consistently rejected.

A second pattern in the Data and Analytics results warrants attention for the ethical dimension of this analysis. A cluster of entities, including Blockchain Technologies LLC, Automation Technologies LLC, Robotic Process Automation LLC, Machine Learning Technologies LLC, Cloud Big Data Technologies LLC, and Artifint Technologies LLC, each show between 500 and 900 total filings with certification rates below 30 percent and denial rates near zero. This pattern of high filing volume, near-zero certification, and near-zero denials indicates that nearly all cases were withdrawn before adjudication. These entities are concentrated in Nebraska, Oklahoma, and Vermont, states with no established technology employment base to support such volumes. This pattern is consistent with fraudulent H-1B filing activity documented in DOL enforcement actions.

How it could inform decision-making. For a General Business candidate, the actionable conclusion from this query is that the top-tier sponsors identified in Query 2 carry essentially zero denial risk. Ernst and Young, Deloitte, PricewaterhouseCoopers, McKinsey, and Goldman Sachs all operate with denial rates below 0.2 percent and certification rates above 95 percent, which represents the most reliable approval profile available in the market. Grant Thornton should be approached with caution despite its standing as a major accounting firm: the combination of denial risk and withdrawal risk produces a certification rate that falls materially below the group average. BCG's withdrawal-driven profile warrants a direct conversation with its immigration team before treating it as a comparably reliable sponsor.

Query 4: Wage Levels and Compliance Gap by Role Group, Certified Cases, 2020–2023

SQL

```
-- QUERY 4A | Wage and Compliance Gap by Role Group, Certified 2020-2023
-- Tables : lca_cases, soc_occupations
-- Technique: LEFT JOIN, CASE WHEN, AVG, conditional SUM
-- ANNUAL_WAGE_FROM and WAGE_COMPLIANCE_GAP are
-- derived columns calculated during ETL normalization
-- =====

WITH role_wages AS (
  SELECT
    CASE
      WHEN LEFT(lc.SOC_CODE, 2) IN ('11','13') THEN 'General Business'
      WHEN LEFT(lc.SOC_CODE, 2) = '15' THEN 'Data & Analytics'
      ELSE NULL
    END
    AS role_group,
    lc.annual_wage_from,
    lc.annual_prevaling_wage,
    lc.wage_compliance_gap,
```

```

        lc.PW_WAGE_LEVEL,
        so.SOC_TITLE
    FROM lca_cases lc
    LEFT JOIN soc_occupations so ON lc.SOC_CODE = so.SOC_CODE
    WHERE lc.FISCAL_YEAR BETWEEN 2020 AND 2023
        AND lc.CASE_STATUS = 'Certified'
        AND LEFT(lc.SOC_CODE, 2) IN ('11','13','15')
        AND lc.annual_wage_from > 0
        AND lc.annual_prevaling_wage > 0
    )
SELECT
    role_group,
    COUNT(*)
    ROUND(AVG(annual_wage_from), 0)
    ROUND(AVG(annual_prevaling_wage), 0)
    ROUND(AVG(wage_compliance_gap), 0)
    ROUND(
        AVG(wage_compliance_gap) / NULLIF(AVG(annual_prevaling_wage), 0) * 100,
        1)
    SUM(CASE WHEN wage_compliance_gap < 0 THEN 1 ELSE 0 END)
    below_prevaling_count,
    ROUND(
        SUM(CASE WHEN wage_compliance_gap < 0 THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*), 0),
        3)
    FROM role_wages
GROUP BY role_group
ORDER BY role_group;

-- QUERY 4B | Wage Level Distribution by Role Group, Certified 2020-2023
-- Tables : lca_cases
-- Technique: CASE WHEN pivot on PW_WAGE_LEVEL
-- Note : PW_WAGE_LEVEL values in this dataset are bare Roman
-- numerals (I / II / III / IV), not "Level I" strings.
-- Confirmed via diagnostic query before fixing.
-- =====

WITH role_levels AS (
    SELECT
        CASE
            WHEN LEFT(SOC_CODE, 2) IN ('11','13') THEN 'General Business'
            WHEN LEFT(SOC_CODE, 2) = '15' THEN 'Data & Analytics'
            ELSE NULL
        END
        AS role_group,
        PW_WAGE_LEVEL
    FROM lca_cases
    WHERE FISCAL_YEAR BETWEEN 2020 AND 2023
        AND CASE_STATUS = 'Certified'
        AND LEFT(SOC_CODE, 2) IN ('11','13','15')
    )
SELECT
    role_group,
    COUNT(*)
    total,
    SUM(CASE WHEN PW_WAGE_LEVEL = 'I' THEN 1 ELSE 0 END)
    level_I,
    SUM(CASE WHEN PW_WAGE_LEVEL = 'II' THEN 1 ELSE 0 END)
    level_II,
    SUM(CASE WHEN PW_WAGE_LEVEL = 'III' THEN 1 ELSE 0 END)
    level_III,
    SUM(CASE WHEN PW_WAGE_LEVEL = 'IV' THEN 1 ELSE 0 END)
    level_IV,
    SUM(CASE WHEN PW_WAGE_LEVEL IS NULL THEN 1 ELSE 0 END)
    level_null,
    ROUND(SUM(CASE WHEN PW_WAGE_LEVEL = 'I' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1)
    pct_I,
    ROUND(SUM(CASE WHEN PW_WAGE_LEVEL = 'II' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1)
    pct_II,
    ROUND(SUM(CASE WHEN PW_WAGE_LEVEL = 'III' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1)
    pct_III,
    ROUND(SUM(CASE WHEN PW_WAGE_LEVEL = 'IV' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1)
    pct_IV
    FROM role_levels
GROUP BY role_group
ORDER BY role_group;

```

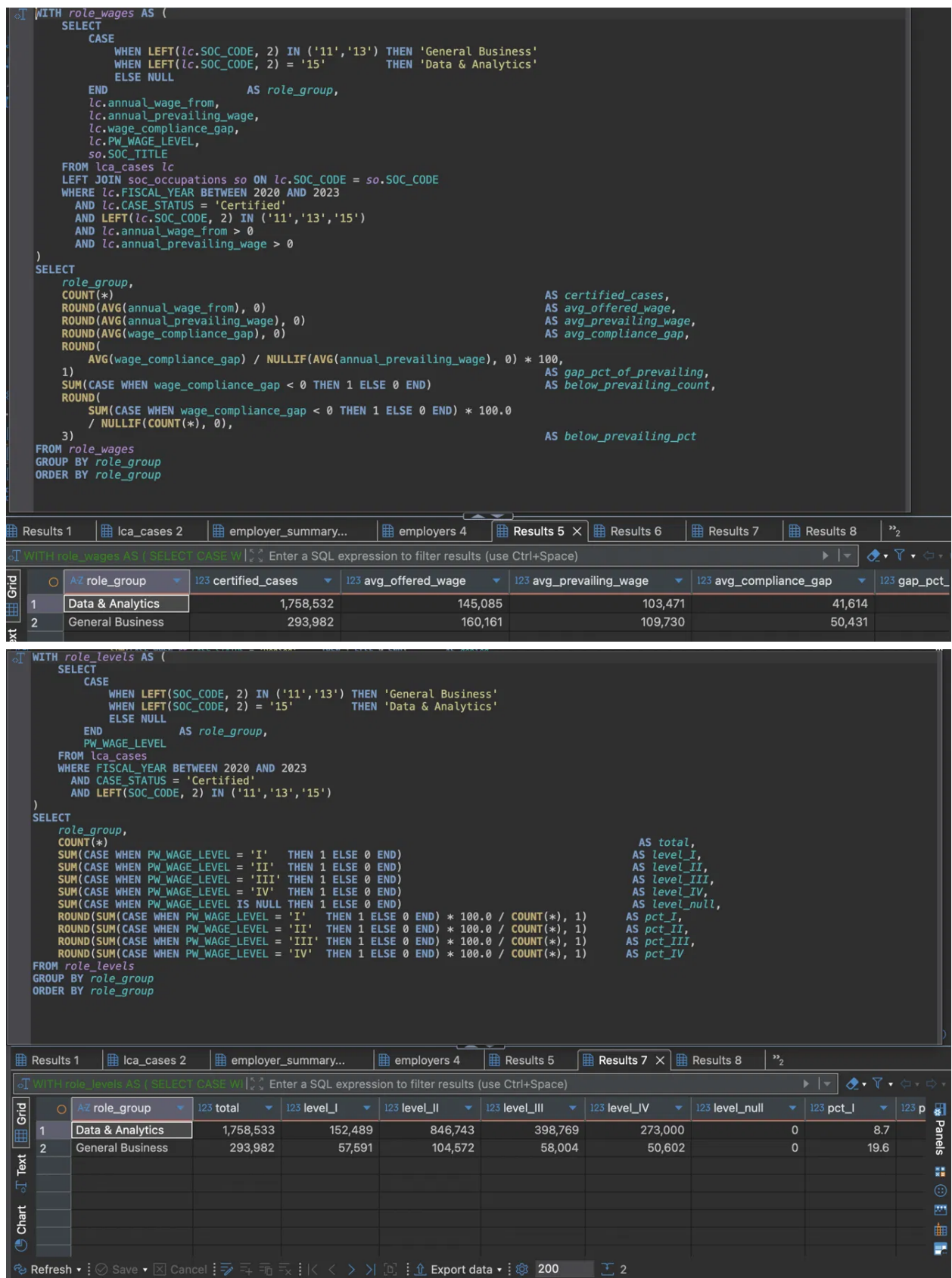


Figure B4. Query 4: Wage Levels and Compliance Gap by Role Group, Certified Cases, 2020–2023 — query result as executed in DBBeaver.

Why it is meaningful. Wage analysis connects the research question to financial outcomes. The offered wage establishes the compensation a candidate can expect. The compliance gap, the difference between the offered wage and the DOL prevailing wage floor, measures how far above the legal minimum employers are actually paying, which is a proxy for genuine market competition for this talent. The prevailing wage level distribution adds a third dimension: whether employers are sponsoring entry-level, mid-career, or senior candidates determines whether an early-career candidate is viable in each track.

What it reveals. General Business roles command higher wages than Data and Analytics roles at every level of analysis. The average offered wage for certified General Business filings across 2020 to 2023 was 160,161 US dollars, compared to 145,085 US dollars for Data and Analytics, a gap of 15,076 US dollars. The prevailing wage floor set by DOL for General Business occupations averaged 109,730 US dollars, against 103,471 US dollars for Data and Analytics, a difference of 6,259 US dollars. After accounting for those different floors, General Business employers still paid 8,817 US dollars more above the prevailing wage threshold than Data and Analytics employers, representing a compliance gap of 46.0 percent versus 40.2 percent of the prevailing wage base. This confirms and extends the wage premium first identified in a prior analysis of Massachusetts-only certified cases: the pattern holds nationally across all case statuses and all five fiscal years.

Both groups show near-zero below-prevailing-wage filing rates. Only 74 Data and Analytics cases and 9 General Business cases recorded a negative wage compliance gap across 1.76 million and 294,000 certified filings respectively. These rates of 0.004 percent and 0.003 percent establish that the H-1B LCA process functions as intended for the role groups studied: the overwhelming majority of employers file at or above the required prevailing wage floor.

The prevailing wage level distribution reveals structurally different sponsor profiles between the two groups. Data and Analytics filings are concentrated at Level II, which accounted for 48.2 percent of all certified cases in that group. Level II represents the mid-career benchmark in the DOL occupational employment survey prevailing wage methodology, specifically a worker who has attained competency in their occupation through experience. General Business filings show a notably higher share of Level I cases at 19.6 percent, compared to 8.7 percent for Data and Analytics, a gap of 10.9 percentage points. Level I defines an entry-level worker not yet fully competent in the occupation. Nearly one in five General Business H-1B filings sponsors an entry-level candidate, a materially higher proportion than in the technology group. The senior share, defined as Levels III and IV combined, is nearly equal across the two groups: 38.2 percent for Data and Analytics and 36.9 percent for General Business.

How it could inform decision-making. The wage and compliance gap results carry two implications for a job seeker. General Business employers pay 46 cents above every dollar of the required floor on average, suggesting genuine market competition for this talent rather than minimum-compliant filing behaviour. The higher Level I share in General Business is the most practically significant finding in this query for an early-career candidate: it establishes that employers in this group actively sponsor candidates at the entry-level stage, meaning a new graduate is not structurally disadvantaged relative to the overall filing pool. In the Data and Analytics group, the Level II concentration indicates that mid-career experience is the dominant credential employers are seeking, which sets a different bar for a candidate without a technical background.

Query 5: Industry Sector Analysis via NAICS Join, All Five Tables, 2020–2023

SQL

```
-- QUERY 5 | Industry Sector Analysis via NAICS JOIN, 2020-2023
-- Tables   : lca_cases, employers, naics_industries
-- Technique: INNER JOIN + LEFT JOIN, CASE WHEN sector mapping
-- Note     : LEFT JOIN on naics_industries preserves employers
--           with 4/5-digit NAICS codes not in the 6-digit
--           Census Bureau lookup table (28.7% coverage gap).
--           naics_unmatched column documents this transparently.
-- =====

WITH industry_tagged AS (
    SELECT
        CASE
            WHEN LEFT(lc.SOC_CODE, 2) IN ('11','13') THEN 'General Business'
            WHEN LEFT(lc.SOC_CODE, 2) = '15'          THEN 'Data & Analytics'
            ELSE NULL
        END AS role_group,
        CASE
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) IN ('5415','5411') THEN 'IT & Computer Services'
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) = '5416'          THEN 'Management Consulting'
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) IN ('5112','5113') THEN 'Software Publishing'
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 2) = '52'           THEN 'Finance & Banking'
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) = '6113'          THEN 'Universities'
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 2) = '62'           THEN 'Healthcare'
            WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 2) = '54'           THEN 'Other Professional
Services'
            ELSE
            END AS industry_sector,
        ni.INDUSTRY_NAME,
        lc.CASE_STATUS,
        lc.annual_wage_from
    FROM lca_cases lc
    JOIN employers e          ON lc.EMPLOYER_ID = e.EMPLOYER_ID
    LEFT JOIN naics_industries ni ON e.NAICS_CODE = ni.NAICS_CODE
    WHERE lc.FISCAL_YEAR BETWEEN 2020 AND 2023
        AND LEFT(lc.SOC_CODE, 2) IN ('11','13','15')
)
SELECT
    role_group,
    industry_sector,
    COUNT(*)
    SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END)
    ROUND(
        SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*), 0), 1
    )
    ROUND(AVG(
        CASE WHEN CASE_STATUS = 'Certified'
            AND annual_wage_from > 0
            THEN annual_wage_from ELSE NULL END
    ), 0)
    avg_wage_certified,
    SUM(CASE WHEN INDUSTRY_NAME IS NULL THEN 1 ELSE 0 END)
FROM industry_tagged
WHERE role_group IS NOT NULL
GROUP BY role_group, industry_sector
ORDER BY role_group, total_filings DESC;
```

WITH industry_tagged AS (
 SELECT
 CASE
 WHEN LEFT(lc.SOC_CODE, 2) IN ('11','13') THEN 'General Business'
 WHEN LEFT(lc.SOC_CODE, 2) = '15' THEN 'Data & Analytics'
 ELSE NULL
 END AS role_group,
 CASE
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) IN ('5415','5411') THEN 'IT & Computer Services'
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) = '5416' THEN 'Management Consulting'
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) IN ('5112','5113') THEN 'Software Publishing'
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 2) = '52' THEN 'Finance & Banking'
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 4) = '6113' THEN 'Universities'
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 2) = '62' THEN 'Healthcare'
 WHEN LEFT(CAST(e.NAICS_CODE AS CHAR), 2) = '54' THEN 'Other Professional Services'
 ELSE 'Other / Unknown'
 END AS industry_sector,
 ni.INDUSTRY_NAME,
 lc.CASE_STATUS,
 lc.annual_wage_from
 FROM lca_cases lc
 JOIN employers e ON lc.EMPLOYER_ID = e.EMPLOYER_ID
 LEFT JOIN naics_industries ni ON e.NAICS_CODE = ni.NAICS_CODE
 WHERE lc.FISCAL_YEAR BETWEEN 2020 AND 2023
 AND LEFT(lc.SOC_CODE, 2) IN ('11','13','15')
)
SELECT
 role_group,
 industry_sector,
 COUNT(*) AS total_filings,
 SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) AS certified,
 ROUND(
 SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) * 100.0
 / NULLIF(COUNT(*), 0), 1
) AS cert_rate_pct,
 ROUND(AVG(
 CASE WHEN CASE_STATUS = 'Certified'
 AND annual_wage_from > 0
 THEN annual_wage_from ELSE NULL END
), 0) AS avg_wage_certified,
 SUM(CASE WHEN INDUSTRY_NAME IS NULL THEN 1 ELSE 0 END) AS naics_unmatched
FROM industry_tagged
WHERE role_group IS NOT NULL
GROUP BY role_group, industry_sector
ORDER BY role_group, total_filings DESC

role_group	industry_sector	total_filings	certified	cert_rate_pct	avg_wage_certified	naics_unmatched
Data & Analytics	Other / Unknown	423,487	393,118	92.8	162,448	
Data & Analytics	Finance & Banking	161,678	151,013	93.4	152,680	
Data & Analytics	Management Consulting	105,412	100,226	95.1	139,177	
Data & Analytics	Other Professional Services	79,570	74,153	93.2	142,662	
Data & Analytics	Software Publishing	78,716	74,671	94.9	157,969	
Data & Analytics	Healthcare	13,653	12,305	90.1	250,190	
Data & Analytics	Universities	12,664	10,222	80.7	86,415	
General Business	Other / Unknown	113,216	103,393	91.3	149,442	

Figure B5. Query 5: Industry Sector Analysis via NAICS Join, All Five Tables, 2020–2023 — query result as executed in DBBeaver.

Why it is meaningful. Employer and role-group findings gain explanatory power when situated within an industry context. A candidate can use industry-level data to identify which sectors concentrate the most H-1B activity for their target role type, which sectors pay the most, and which produce the most reliable approval outcomes. This query is also the only one in the set that joins all five tables simultaneously, demonstrating the full relational structure of the lca_h1b schema and the analytical reach of the normalised design.

A methodological note applies to the NAICS join. The naics_industries reference table uses six-digit Census Bureau codes, while 28.7 percent of employer records carry four-digit or five-digit NAICS codes that have no matching entry in the reference. A LEFT JOIN is used throughout to preserve all employer records. The Software Publishing sector is absent from the Census Bureau six-digit lookup and therefore shows a 100 percent non-match rate in the industry name column. The filing counts and wage figures for that sector are derived from the filing data and are unaffected by this lookup gap.

What it reveals. IT and Computer Services is the dominant sector for Data and Analytics H-1B filings, accounting for 1,014,729 cases or 53.7 percent of all Data and Analytics filings in the period. General Business filings are distributed more evenly across sectors, with no single industry exceeding 113,216 filings. This structural difference reflects the underlying

economics: technology firms sponsor H-1B workers at industrial scale for software and IT roles, while General Business sponsorship is spread across consulting, finance, professional services, and technology employers hiring for business functions.

The wage premium for General Business over Data and Analytics holds within every high-volume commercial sector and is not a compositional artefact. Within IT and Computer Services, General Business roles averaged 177,370 US dollars versus 135,691 US dollars for Data and Analytics, a gap of 41,679 US dollars. Within Finance and Banking, the gap widens to 54,670 US dollars, with General Business averaging 207,350 US dollars against 152,680 US dollars for Data and Analytics. Within Management Consulting, the gap narrows to 10,264 US dollars. This rules out the possibility that the overall wage premium in Query 4 simply reflected General Business being more concentrated in high-paying industries.

Healthcare is the one documented exception to the General Business premium. In that sector, Data and Analytics roles averaged 250,190 US dollars against 216,311 US dollars for General Business, reversing the pattern seen elsewhere. Healthcare also carries low filing volume and below-average certification in both groups, at 90.1 percent for Data and Analytics and 89.0 percent for General Business. Finance and Banking is the highest-paying high-volume sector for General Business roles at 207,350 US dollars average, with a 90.8 percent certification rate across 56,056 filings. Management Consulting achieves the highest General Business certification rate at 93.8 percent alongside wages of 149,441 US dollars across 27,459 filings. Universities represent the lowest-wage and lowest-certification sector in both groups: Data and Analytics university roles averaged 86,415 US dollars at an 80.7 percent certification rate, and General Business university roles averaged 113,813 US dollars at 82.9 percent.

How it could inform decision-making. For a General Business candidate, the industry analysis produces a concrete strategic conclusion. Finance and Banking is the highest-paying high-volume sector at 207,350 US dollars average with 90.8 percent certification across 56,056 filings, a combination of volume, wage, and approval reliability that makes it the most attractive industry target. IT and Computer Services follows at 177,370 US dollars with 92.9 percent certification and 44,085 filings, confirming that technology employers sponsoring business-function roles do so at competitive wages and high approval rates. Management Consulting offers the most reliable approval environment at 93.8 percent certification alongside wages of 149,441 US dollars. Universities should be deprioritised on every measurable dimension: fewer filings, substantially lower wages, and lower certification rates than any commercial sector.

Query 6: Year-over-Year Sponsorship Trend, LAG Window Function, 2020–2023

SQL

```
-- QUERY 6 | Year-over-Year Sponsorship Trend, 2020-2023
-- Tables   : lca_cases
-- Technique: CTE aggregation + LAG() window function
--          2020 rows show NULL for LAG columns (no prior year)
-- =====

WITH annual_summary AS (
  SELECT
    CASE
      WHEN LEFT(SOC_CODE, 2) IN ('11','13') THEN 'General Business'
      WHEN LEFT(SOC_CODE, 2) = '15'         THEN 'Data & Analytics'
      ELSE NULL
    
```

```

        END
        FISCAL_YEAR,
        COUNT(*)
        SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END)
        ROUND(
            SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) * 100.0
            / NULLIF(COUNT(*), 0), 1
        )
        ROUND(AVG(
            CASE WHEN CASE_STATUS = 'Certified' AND annual_wage_from > 0
            THEN annual_wage_from ELSE NULL END
        ), 0)
        FROM lca_cases
        WHERE FISCAL_YEAR BETWEEN 2020 AND 2023
        AND LEFT(SOC_CODE, 2) IN ('11','13','15')
        GROUP BY role_group, FISCAL_YEAR
    ),
    with_lag AS (
        SELECT
            role_group,
            FISCAL_YEAR,
            total_filings,
            certified,
            cert_rate_pct,
            avg_wage_certified,
            LAG(total_filings) OVER (PARTITION BY role_group ORDER BY FISCAL_YEAR) AS prev_total,
            LAG(certified) OVER (PARTITION BY role_group ORDER BY FISCAL_YEAR) AS prev_certified,
            LAG(avg_wage_certified) OVER (PARTITION BY role_group ORDER BY FISCAL_YEAR) AS prev_avg_wage
        FROM annual_summary
        WHERE role_group IS NOT NULL
    )
    SELECT
        role_group,
        FISCAL_YEAR,
        total_filings,
        certified,
        cert_rate_pct,
        avg_wage_certified,
        prev_total,
        total_filings - prev_total
        ROUND(
            (total_filings - prev_total) * 100.0 / NULLIF(prev_total, 0), 1
        )
        avg_wage_certified - prev_avg_wage
        ROUND(
            (avg_wage_certified - prev_avg_wage) * 100.0
            / NULLIF(prev_avg_wage, 0), 1
        )
        FROM with_lag
        ORDER BY role_group, FISCAL_YEAR;

```

```

AS role_group,
AS total_filings,
AS certified,
AS cert_rate_pct,
AS avg_wage_certified
AS filing_change_abs,
AS filing_change_pct,
AS wage_change_abs,
AS wage_change_pct

```

The screenshot displays a SQL query in DBeaver's editor, which is then executed to show a results table. The query is a multi-part SQL statement using Common Table Expressions (CTEs) and window functions to calculate year-over-year trends for sponsorship filings. The results table shows data for two role groups: 'Data & Analytics' and 'General Business', across fiscal years 2020 to 2023. The columns include 'role_group', 'FISCAL_YEAR', 'total_filings', 'certified', 'cert_rate_pct', 'avg_wage_certified', and 'prev_t' (previous year's total filings). The 'Data & Analytics' group shows a decline in total filings from 496,327 in 2020 to 466,142 in 2023. The 'General Business' group shows an increase in total filings from 76,805 in 2020 to 83,780 in 2023.

	role_group	FISCAL_YEAR	total_filings	certified	cert_rate_pct	avg_wage_certified	prev_t
1	Data & Analytics	2,020	496,327	457,413	92.2	131,706	
2	Data & Analytics	2,021	472,775	432,045	91.4	118,277	
3	Data & Analytics	2,022	454,665	424,864	93.4	144,996	
4	Data & Analytics	2,023	466,142	444,211	95.3	185,021	
5	General Business	2,020	76,805	70,267	91.5	139,621	
6	General Business	2,021	81,126	74,058	91.3	134,896	
7	General Business	2,022	77,530	71,132	91.7	164,526	
8	General Business	2,023	83,780	78,525	93.7	198,416	

Figure B6. Query 6: Year-over-Year Sponsorship Trend, LAG Window Function, 2020–2023 — query result as executed in DBeaver.

Why it is meaningful. Aggregate totals establish the size of the market for each role group. Year-over-year change rates establish its direction. The LAG window function surfaces the directional signal year by year, revealing whether trends are accelerating, decelerating, or reversing. The distinction between a temporary disruption and a secular shift in employer behaviour is not visible in a single-year aggregate and requires multi-year sequential comparison. This query also demonstrates the use of a window function applied to an aggregated result set, a technique that requires the two-CTE structure used here and could not be achieved with a simple GROUP BY.

What it reveals. The LAG window function surfaces a structural divergence between the two groups that the aggregate numbers in Query 1 obscured: General Business sponsorship volume grew 9.1 percent net over the four-year period while Data and Analytics volume declined 6.1 percent. Data and Analytics filings peaked at 496,327 in 2020 and never recovered to that level, ending at 466,142 in 2023 after declining in both 2021 and 2022. General Business filings grew from 76,805 in 2020 to 83,780 in 2023, absorbing a mid-period dip in 2022 and recovering more strongly in 2023 at 8.1 percent compared to 2.5 percent for Data and Analytics. These trends indicate that General Business sponsorship is on an upward secular trajectory while Data and Analytics sponsorship has entered a post-peak consolidation phase, consistent with the technology sector hiring correction that began in late 2022.

The COVID-19 shock of 2021 produced a double hit on Data and Analytics: simultaneous volume and wage compression. Data and Analytics filings fell 4.7 percent in 2021 while average certified wages dropped 10.2 percent, from 131,706 US dollars to 118,277 US dollars. General Business filings grew 5.6 percent in the same year while wages declined only 3.4 percent, from 139,621 US dollars to 134,896 US dollars. Technology employers responded to pandemic uncertainty by reducing H-1B activity and compressing wages simultaneously, while business-function employers expanded hiring and largely protected compensation.

Both groups underwent extraordinary wage growth in 2022 and 2023, driven by post-pandemic labour market tightening. Data and Analytics wages rose 22.6 percent in 2022 and a further 27.6 percent in 2023, reaching 185,021 US dollars, a 40.5 percent increase from the 2020 baseline. General Business wages rose 22.0 percent in 2022 and 20.6 percent in 2023, reaching 198,416 US dollars, a 42.1 percent increase from 2020. The General Business wage premium fluctuated across the period: 7,915 US dollars in 2020, widening to a peak of 19,530 US dollars in 2022, then narrowing to 13,395 US dollars in 2023 as Data and Analytics wages grew proportionally faster in recovery. The premium is structural rather than cyclical: it persisted and fluctuated but did not disappear through a full economic cycle.

How it could inform decision-making. A candidate evaluating these two tracks in 2024 should weight the following observations. General Business H-1B sponsorship volume has grown through the study period and recovered fastest in 2023, indicating an upward trajectory. Data and Analytics volume peaked in 2020 and has not returned to that level, coinciding with the technology sector's documented reduction in workforce activity. Wage growth has been equivalent across both groups in percentage terms, meaning the General Business premium is maintained structurally rather than widening or closing directionally. The 2023 wage levels, 198,416 US dollars for General Business and 185,021 US dollars for Data and Analytics, represent the most current market signal available in this dataset, and both figures are materially higher than any prior year in the study period.

Query 7: Durable H-1B Sponsors Across All Four Complete Years, 2020–2023

SQL

```
-- QUERY 7 | Durable H-1B Sponsors Present in All 4 Years, 2020-2023
-- Tables   : lca_cases, employers, worksites (all five tables used)
-- Technique: 3-CTE chain, HAVING COUNT(DISTINCT FISCAL_YEAR) = 4
--           to enforce continuity, ROW_NUMBER() window function
--           to identify dominant worksite state per employer
-- =====

WITH role_filings AS (
  SELECT
    lc.EMPLOYER_ID,
    lc.WORKSITE_ID,
    CASE
      WHEN LEFT(lc.SOC_CODE, 2) IN ('11','13') THEN 'General Business'
      WHEN LEFT(lc.SOC_CODE, 2) = '15'         THEN 'Data & Analytics'
      ELSE NULL
    END
    AS role_group,
    lc.CASE_STATUS,
    lc.annual_wage_from,
    lc.FISCAL_YEAR
  FROM lca_cases lc
  WHERE lc.FISCAL_YEAR BETWEEN 2020 AND 2023
        AND LEFT(lc.SOC_CODE, 2) IN ('11','13','15')
),
durable_employers AS (
```

```

SELECT
    role_group,
    EMPLOYER_ID,
    COUNT(*)
    SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END)
    COUNT(DISTINCT FISCAL_YEAR)
    ROUND(
        SUM(CASE WHEN CASE_STATUS = 'Certified' THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*), 0), 1
    )
    ROUND(AVG(
        CASE WHEN CASE_STATUS = 'Certified' AND annual_wage_from > 0
        THEN annual_wage_from ELSE NULL END
    ), 0)
FROM role_filings
WHERE role_group IS NOT NULL
GROUP BY role_group, EMPLOYER_ID
HAVING COUNT(DISTINCT FISCAL_YEAR) = 4
AND COUNT(*) >= 200
),
dominant_worksite AS (
    SELECT
        rf.EMPLOYER_ID,
        rf.role_group,
        ws.WORKSITE_STATE,
        ROW_NUMBER() OVER (
            PARTITION BY rf.EMPLOYER_ID, rf.role_group
            ORDER BY COUNT(*) DESC
        )
        AS rn
    FROM role_filings rf
    JOIN worksites ws ON rf.WORKSITE_ID = ws.WORKSITE_ID
    WHERE rf.role_group IS NOT NULL
    GROUP BY rf.EMPLOYER_ID, rf.role_group, ws.WORKSITE_STATE
)
SELECT
    de.role_group,
    e.EMPLOYER_NAME,
    e.EMPLOYER_STATE,
    dw.WORKSITE_STATE
    de.total_filings,
    de.certified,
    de.cert_rate_pct,
    de.avg_wage,
    de.years_active
    AS top_worksite_state,
FROM durable_employers de
JOIN employers e ON de.EMPLOYER_ID = e.EMPLOYER_ID
LEFT JOIN dominant_worksite dw
    ON de.EMPLOYER_ID = dw.EMPLOYER_ID
    AND de.role_group = dw.role_group
    AND dw.rn = 1
ORDER BY de.role_group, de.certified DESC;

```

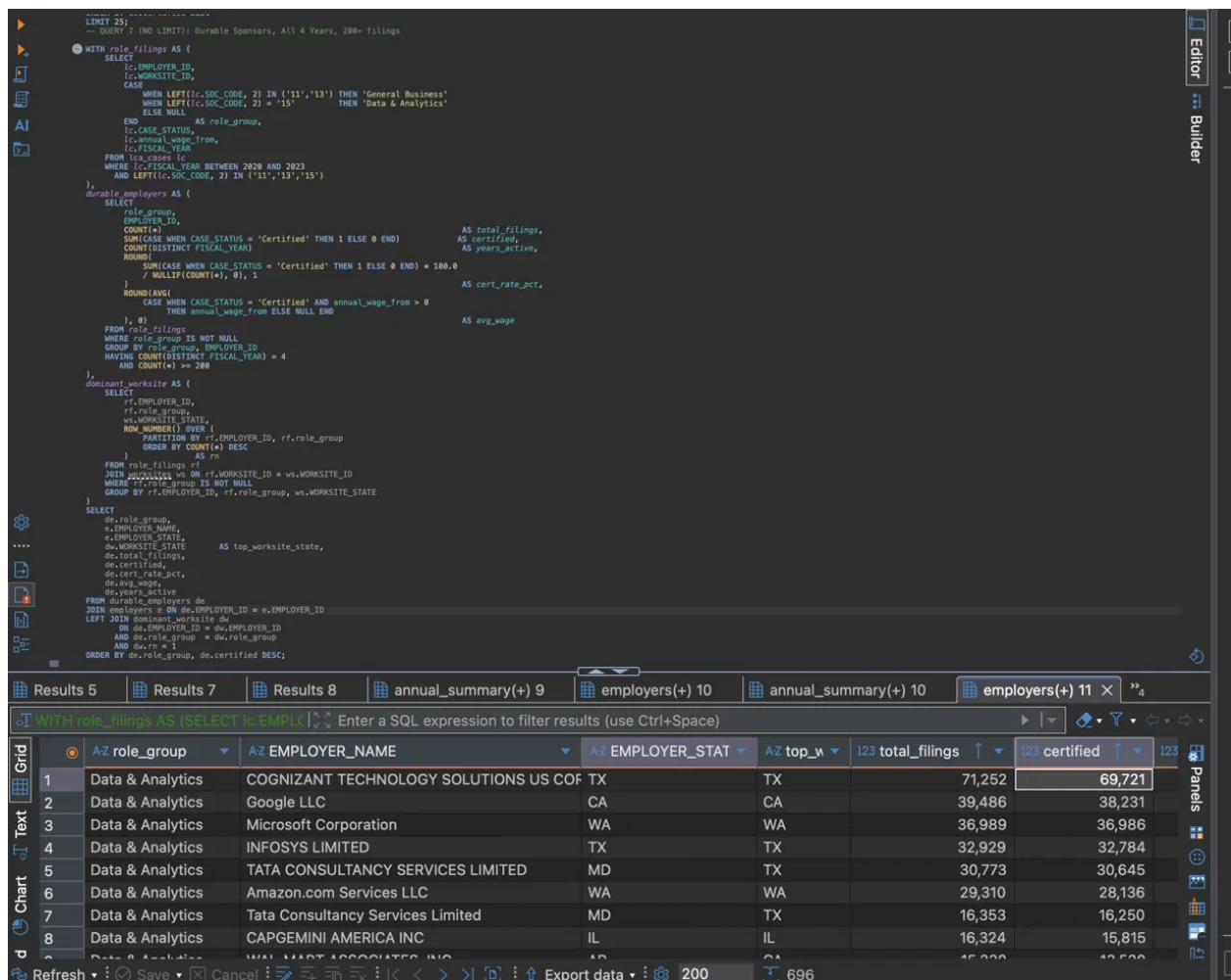


Figure B7. Query 7: Durable H-1B Sponsors Across All Four Complete Years, 2020–2023 — query result as executed in DBeaver.

Why it is meaningful. Filing volume in a single year does not establish sponsorship reliability. An employer who filed heavily in 2021 but withdrew from the market in 2022 is not a viable long-term target. The durability filter identifies employers who sponsored H-1B workers in every year from 2020 to 2023, through the pandemic, the post-COVID hiring surge, and the technology sector correction, demonstrating that their commitment to this channel is not contingent on market conditions. The 200-filing floor removes employers whose annual presence is too limited to indicate a structural programme rather than an occasional filing.

What it reveals. The durability filter produces a dramatically smaller pool for General Business than for Data and Analytics: 77 distinct employer entries versus 619. This asymmetry does not mean General Business sponsorship is less reliable. It reflects the structural difference in market concentration: General Business H-1B sponsorship is held within a small number of very large, very consistent sponsors, while Data and Analytics sponsorship is distributed across a far wider employer ecosystem at varying scales.

Ernst and Young is the most durable General Business H-1B sponsor in the national dataset. Across all four complete years, Ernst and Young certified 13,329 General Business filings at a 97.8 percent certification rate, with workers concentrated in New York. Deloitte's multiple legal entities collectively certified approximately 12,589 General Business cases, making it the effective second-largest durable sponsor when entities are consolidated. Microsoft Corporation is the only employer in either group to certify 100 percent of all filings across all

four years in both role groups simultaneously. Amazon, Google, PricewaterhouseCoopers, Apple, Infosys, KPMG, Goldman Sachs, and Citibank also appear on the durable General Business list with certification rates above 92 percent.

The geographic concentration of durable General Business sponsors is notable. Of 77 durable General Business employer entries, 30 concentrated their workers primarily in California and 26 in New York. General Business H-1B sponsorship at scale is therefore concentrated in two states, reflecting the dominance of professional services, finance, and technology employer headquarters in California and New York.

BCG is present in the durable General Business list with the lowest certification rate of any employer in that group: 71.2 percent across 911 total filings and 649 certified cases, with the dominant worksite in New York. The average certified wage of 190,963 US dollars is among the highest in the group, confirming that BCG's non-certification is not a wage compliance issue. BCG's pattern of high-value, high-withdrawal filings persists across all four years, indicating a structural feature of its immigration practice rather than a temporary anomaly. Several Data and Analytics employers in this query show average wage figures above 500,000 US dollars. These figures are averages distorted by extreme outlier filings within small certified-case counts and are not representative wage values. They are excluded from all wage comparisons in this report.

How it could inform decision-making. The durability filter produces the most operationally useful output in this report for a job seeker. An employer who sponsored H-1B filings in every year from 2020 to 2023 has demonstrated a structural commitment to this channel that is not contingent on market conditions. The General Business durable list is short enough to constitute a concrete target list: Ernst and Young, Deloitte, PricewaterhouseCoopers, Google, Microsoft, Amazon, Apple, Infosys, KPMG, Goldman Sachs, Citibank, and McKinsey all appear with four-year durability and certification rates above 92 percent. A candidate approaching any of these employers can reasonably expect that the immigration infrastructure, internal legal support, and institutional willingness to sponsor are already established. BCG remains on the durable list but with a documented certification pattern that warrants direct inquiry before treating it as equivalent to the Big Four or technology giants.

Part C: Ethical Reflection

c.1 The Dataset's Disclosure Mandate Does Not Extend to Commercial Use

DOL regulations implementing Section 212(n) of the Immigration and Nationality Act require that certified labor condition applications be made publicly available (Immigration and Nationality Act of 1952). The disclosure obligation exists to enable wage compliance accountability, not to create a commercial data asset. That regulatory origin establishes the foundational ethical boundary for this dataset: uses aligned with accountability are defensible; uses that repurpose the data for commercial surveillance, competitive intelligence, or individual targeting are not.

Edquist et al. argue that ethical data use must offer a clear benefit to both the organization and the individuals whose data is involved (2022). They further hold that those individuals should not be surprised by how their data is used. H-1B workers whose occupational and wage data appears in this dataset did not consent to commercial use of that information. That absence of consent is the primary ethical constraint on any downstream application built from LCA records.

c.2 Legitimate Uses Are Grounded in the Accountability Purpose the Dataset Was Created to Serve

The clearest ethical use of LCA data is the one for which it was created: verifying that employers meet prevailing wage obligations under the H-1B program. This analysis identified 74 Data and Analytics cases and nine General Business cases that carried a negative wage compliance gap while remaining in certified status, meaning the offered wage fell below the legally required prevailing wage floor. Wage compliance monitoring conducted by DOL researchers, labor advocates, or academic institutions is an unambiguous ethical application. The dataset exists precisely to make such monitoring possible, and the near-zero violation rate it surfaces is itself a meaningful regulatory signal.

A second ethical use is redressing information asymmetry for visa-dependent workers. International candidates face structural disadvantage in assessing which employers will reliably sponsor H-1B petitions. This analysis provides a durability-filtered, denial-rate-adjusted view of employer behavior across 3.47 million cases and four complete fiscal years. The data was disclosed by employers to a public body. Making that record legible to the workers those employers seek to hire is consistent with the program's protective intent.

A third defensible use is public-interest fraud research. Query 3 surfaced a cluster of entities — including Blockchain Technologies LLC, Automation Technologies LLC, Robotic Process Automation LLC, and Machine Learning Technologies LLC — that each filed 500 to 900 cases between 2020 and 2023 with certification rates between 25 and 50 percent, zero denial rates, and near-total withdrawal rates, concentrated in Nebraska, Oklahoma, and Vermont. That pattern is consistent with fraudulent filing activity described in DOL Office of Foreign Labor Certification enforcement actions, where shell employers file large application volumes with no intent to employ and withdraw before adjudication. Data analysts who surface such patterns and refer findings to enforcement bodies are engaging in a legitimate protective function.

c.3 Employer Surveillance Converts Regulatory Disclosure into a Competitive Weapon

The same wage and worksite fields that enable compliance monitoring create a near-complete map of each employer's hiring activity by role, location, and compensation band. A competitor with database access could systematically identify which roles a rival is filing for, at which worksites, and at what offered wage level — enabling targeted talent acquisition or strategic wage undercutting. This use violates the consent principle directly. The employer disclosed its hiring data to satisfy a regulatory obligation, not to provide commercial intelligence to market rivals. Any organization that aggregates and monetizes LCA records for talent intelligence products should be required to publish a plain-language disclosure of that commercial purpose and provide employers with a documented mechanism to flag data quality errors about their own records.

c.4 Aggregated Records Can Expose Individual Workers Without Naming Them

LCA filings do not identify individual H-1B workers by name. However, at small employers, the combination of employer name, SOC code, worksite state, and prevailing wage level is sufficient to narrow to one or two individuals. A worker employed as the sole Management Analyst at a 50-person firm in a specific state at a specific wage level is effectively identifiable from public records without any direct naming. That re-identification risk is structural, not speculative. DOL does not apply data minimization to the public disclosure file. Any organization that builds a secondary product on LCA data should assess re-identification risk for small-employer records before publication and suppress worksite-level specificity where the combination of fields creates a meaningful risk of individual identification.

c.5 Algorithmic Misrepresentation Harms the Workers the Data Is Meant to Serve

Sponsorship volume and certification rate data could be incorporated into scoring tools that rank employers for international job seekers without disclosing the methodology or its limitations. Query 7 identified 619 durable Data and Analytics employers and 77 durable General Business employers meeting the four-year durability threshold with 200 or more filings. An application that ranks Cognizant first on the basis of 69,722 certified cases would mislead a candidate who does not know that Cognizant's volume reflects IT staffing and contract placement rather than direct employment. The score would be technically accurate and substantively misleading at the same time. Edquist et al. argue that ethical data use requires offering consumers some measure of agency, which is impossible without disclosure of how a score is constructed and what it does not measure (2022). A scoring tool built on LCA data must disclose, at minimum, that volume does not predict individual approval probability, that IT staffing employers follow a different sponsorship model than direct employers, and that denial rates in this dataset reflect LCA adjudication outcomes rather than USCIS petition outcomes.

c.6 Structural Bias in SOC Classification Propagates into Any Downstream Model

The Standard Occupational Classification system used in LCA filings was not designed to reliably distinguish general business track roles from technical roles. Business Operations Specialists, Management Analysts, Financial Analysts, and Market Research Analysts each

map to different SOC codes despite overlapping functions, and an employer who files a business-track role under a non-standard code does not appear in any General Business count in this analysis. Query 5 documented a 28.7 percent NAICS code mismatch rate between employer records and the Census six-digit lookup table, confirming that structured taxonomies in this dataset systematically fail to capture full population coverage. Any predictive or advisory model trained on LCA data inherits that undercount. A recruiting algorithm trained on these records would systematically underestimate the scale of the general-business H-1B market and steer candidates away from a segment of the labor market that the data incompletely represents. Edquist et al. report that only 17 percent of surveyed companies had a dedicated data governance committee that includes risk and legal professionals, a structural gap that leaves most organizations without the oversight capacity to detect classification biases before they enter production models (2022).

c.7 Policy Recommendations for Responsible Organizational Use

A recruitment analytics firm, career platform, or HR analytics team that incorporates LCA data should adopt five concrete practices.

The organization should publish a plain-language methodology disclosure for any product that surfaces sponsorship scores or employer rankings. The disclosure should specify which data years are included, how the role classification was constructed, what the score measures, and what it does not predict. Transparency about methodology is the minimum standard for any tool that influences candidate behavior.

The organization should apply a data minimization protocol to small-employer records. Employer records with fewer than 50 total LCA filings in a given year should have worksite city and county fields suppressed in any published output, materially reducing the risk of re-identifying individual workers from the combination of public fields.

The organization should implement structured entity resolution before any employer ranking is published. Query 3 identified HCL America Solutions Inc as a distinct legal entity from HCL America Inc, a distinction carrying a 20-percentage-point difference in denial rates that a naive name-match would collapse into a single misleading figure. Entity resolution logic should be documented, versioned, and subject to periodic audit.

The organization should establish an annual data retention review for historical LCA records. Records older than seven years should be archived and excluded from active scoring models. Labor market conditions change materially over longer windows. Denial rates from pre-pandemic years do not reliably predict post-pandemic approval outcomes.

The organization should designate a cross-functional data ethics owner with authority to review new product uses of LCA data before launch. Edquist et al. document that ethical obligation extends beyond legal compliance and requires business unit leaders to vet data strategies jointly with legal and risk teams (2022). A mandatory pre-launch ethics review operationalizes that principle and ensures that each new use of LCA data is evaluated against the dataset's regulatory purpose, not only against technical feasibility.

References

- Bian, Z. (2024). H-1B LCA disclosure data 2020–2024 [Data set]. Kaggle.
<https://www.kaggle.com/datasets/zongaobian/h1b-lca-disclosure-data-2020-2024>
- Edquist, A., Grennan, L., Griffiths, S., & Rowshankish, K. (2022, September 23). Data ethics: What it means and what it takes. McKinsey & Company.
<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/data-ethics-what-it-means-and-what-it-takes>
- Immigration and Nationality Act of 1952, Pub. L. No. 82–414, 66 Stat. 163 (codified as amended at 8 U.S.C. § 1182(n)).
- U.S. Census Bureau. (2022). 2022 NAICS six-digit codes [Data set].
https://www.census.gov/naics/2022NAICS/6-digit_2022_Codes.xlsx
- U.S. Department of Homeland Security. (2024). STEM OPT hub. Study in the States.
<https://studyinthestates.dhs.gov/stem-opt-hub>
- U.S. Department of Labor, Employment and Training Administration. (2024). LCA programs (H-1B, H-1B1, E-3) disclosure data, FY2020–FY2024 [Data set]. Office of Foreign Labor Certification. <https://www.dol.gov/agencies/eta/foreign-labor/performance>
- U.S. Department of Labor, Employment and Training Administration. (2024). Office of Foreign Labor Certification. <https://www.dol.gov/agencies/eta/foreign-labor>